UC SAN DIEGO / ECE 251C

5

Gibbs phenomena and the need for localization

Executive summary

Fourier partial sums are orthogonal projections and therefore give the best approximation using a prescribed set of frequencies. Nevertheless, near a jump, their ringing persists even as the energy error tends to zero. We derive this Gibbs phenomenon for a step function and use the Dirichlet kernel to explain its origin. Fejér averaging removes the overshoot by changing the Fourier weights. We then change the representation itself: normalized indicators give local averages, and Haar wavelets describe local differences. The same projection principle thus leads from global frequency content to localized changes at different scales.

5.1 Fourier analysis on a finite interval

Our discussion of sampling used Fourier series to describe periodic signals through their frequencies. Here we regard one period as a function on $[0,1]$ and examine how a finite number of those frequencies approximates it. This requires a notion of approximation error, so we work in the finite-energy signal space

$$L^2[0,1] := \left\{ f : [0,1] \to \mathbb{C} : \int_0^1 |f(t)|^2 \, \mathrm{d}t < \infty \right\}.$$

The inner product induces the norm:

$$\langle f, g \rangle_{L^2} = \int_0^1 f(t) g^*(t) \, \mathrm{d}t, \qquad \|f\|_{L^2} = \sqrt{\langle f, f \rangle_{L^2}}. \tag{5.1}$$

As in the continuous-domain setting on $\mathbb{R}$, the inner product is linear in its first argument and conjugates its second argument. In particular, $f(t)f^*(t) = |f(t)|^2$, so the inner product induces the energy norm. We identify functions that agree except on a set of measure zero; in particular, changing endpoint values does not affect an element of $L^2[0,1]$.

Period-one exponentials. For $\mathrm{e}^{\mathrm{j}\omega t}$ to repeat after one unit, we need

$$\mathrm{e}^{\mathrm{j}\omega(t+1)} = \mathrm{e}^{\mathrm{j}\omega t} \quad \Longleftrightarrow \quad \mathrm{e}^{\mathrm{j}\omega} = 1 \quad \Longleftrightarrow \quad \omega = 2\pi k, \ k \in \mathbb{Z}.$$

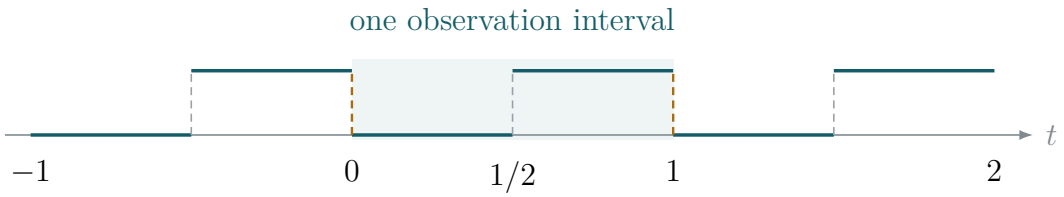


Figure 5.1: A step on $[0, 1]$, repeated periodically. The endpoint mismatch creates a second jump where consecutive periods meet.

Define $e_k(t) := \mathrm{e}^{\mathrm{j}2\pi kt}$. Then $e_k(t+1) = e_k(t)$. Here 1 is a period, not necessarily the *smallest* period; $e_0 = 1$ is constant.

No endpoint-matching condition is required for membership in $L^2[0, 1]$. Periodicity enters through the representing functions: extending a Fourier series beyond the observed interval repeats the signal with period one. A mismatch between the one-sided endpoint limits therefore becomes a jump in this periodic extension.

5.2 Fourier coefficients are inner products

The exponentials provide coordinates for signals on the interval. We first check the geometric property that makes those coordinates easy to compute: orthonormality.

Definition. Two functions are *orthogonal* if their inner product is zero. A family is *orthonormal* if distinct members are orthogonal and every member has norm one.

The Fourier functions $\{e_k\}_{k\in\mathbb{Z}}$ are orthonormal. For integers k, ℓ, we calculate

$$\begin{aligned}\langle e_k, e_\ell\rangle_{L^2} &= \int_0^1 \mathrm{e}^{\mathrm{j}2\pi kt}\mathrm{e}^{-\mathrm{j}2\pi\ell t}\,\mathrm{d}t = \int_0^1 \mathrm{e}^{\mathrm{j}2\pi(k-\ell)t}\,\mathrm{d}t \\ &= \begin{cases} 1, & k = \ell, \\ \left[\dfrac{\mathrm{e}^{\mathrm{j}2\pi(k-\ell)t}}{\mathrm{j}2\pi(k-\ell)}\right]_0^1 = 0, & k \neq \ell. \end{cases}\end{aligned}$$

Thus $\langle e_k, e_\ell\rangle_{L^2} = \delta[k-\ell]$. In particular, $\|e_k\|_{L^2} = 1$.

Recall. For a T-periodic signal, the Fourier frequencies are $k\omega_0$, where $\omega_0 = 2\pi/T$, and

$$c_k = \frac{1}{T}\int_{t_0}^{t_0+T} f(t)\mathrm{e}^{-\mathrm{j}k\omega_0 t}\,\mathrm{d}t.$$

For the interval under consideration, $T = 1$, so no additional normalization factor is needed. The Fourier coefficients are precisely the inner products with the normalized exponentials.

Definition. The Fourier coefficients and the Nth partial sum are

$$c_k \coloneqq \langle f, e_k \rangle_{L^2} = \int_0^1 f(t) \mathrm{e}^{-\mathrm{j}2\pi kt} \, \mathrm{d}t, \tag{5.2}$$

$$S_N f \coloneqq \sum_{k=-N}^{N} c_k e_k, \qquad N = 0, 1, 2, \ldots. \tag{5.3}$$

The distinction between analysis and reconstruction is important: analysis pairs f with e_k^*, whereas reconstruction multiplies the resulting coefficient by e_k. The opposite exponential signs are a consequence of the inner-product convention.

The coefficients are well defined. The elementary inequality

$$(|f(t)| - 1)^2 \geq 0 \quad \Longrightarrow \quad 2|f(t)| \leq 1 + |f(t)|^2$$

gives

$$2 \int_0^1 |f(t)| \, \mathrm{d}t \leq 1 + \int_0^1 |f(t)|^2 \, \mathrm{d}t < \infty.$$

Since $|e_k| = 1$, the absolute value of the integrand in equation (5.2) is $|f|$. Every coefficient is therefore well defined.

Extracting the coordinates. To see why the inner products give the coefficients, suppose first that $f = \sum_{k=-N}^{N} a_k e_k$. For $|\ell| \leq N$,

$$\langle f, e_\ell \rangle_{L^2} = \sum_{k=-N}^{N} a_k \langle e_k, e_\ell \rangle_{L^2} = \sum_{k=-N}^{N} a_k \delta[k - \ell] = a_\ell.$$

Orthogonality removes every term except the one indexed by ℓ. Thus the inner product extracts the coordinate of f along the chosen basis function, exactly as a dot product extracts a coordinate in an orthonormal basis of a finite-dimensional vector space.

Example. Find the coefficients of

$$f(t) = 3 + 2\cos(2\pi t) - \sin(4\pi t).$$

Using $2\cos\theta = \mathrm{e}^{\mathrm{j}\theta} + \mathrm{e}^{-\mathrm{j}\theta}$ and $\sin\theta = (\mathrm{e}^{\mathrm{j}\theta} - \mathrm{e}^{-\mathrm{j}\theta})/(2\mathrm{j})$,

$$\begin{aligned} f(t) &= 3 + \mathrm{e}^{\mathrm{j}2\pi t} + \mathrm{e}^{-\mathrm{j}2\pi t} - \frac{\mathrm{e}^{\mathrm{j}4\pi t} - \mathrm{e}^{-\mathrm{j}4\pi t}}{2\mathrm{j}} \\ &= 3e_0(t) + e_1(t) + e_{-1}(t) + \frac{\mathrm{j}}{2} e_2(t) - \frac{\mathrm{j}}{2} e_{-2}(t). \end{aligned}$$

Here $1/\mathrm{j} = -\mathrm{j}$. Reading off the coordinates gives

$$c_0 = 3, \qquad c_1 = c_{-1} = 1, \qquad c_2 = \frac{\mathrm{j}}{2}, \quad c_{-2} = -\frac{\mathrm{j}}{2};$$

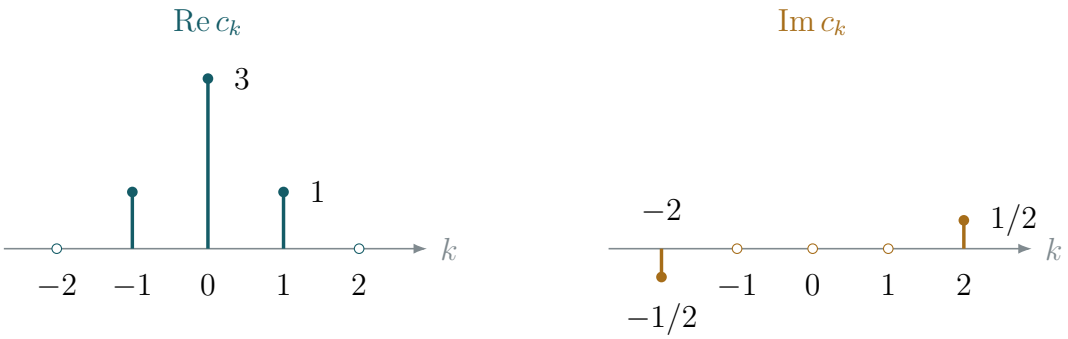


Figure 5.2: The real part of this spectrum is even; the imaginary part is odd. The coefficient at $k = -2$ is the conjugate of the coefficient at $k = 2$.

all other coefficients are zero.

For any real-valued f, conjugating the coefficient integral gives

$$c_k^* = \int_0^1 f(t) \mathrm{e}^{\mathrm{j}2\pi kt} \, \mathrm{d}t = c_{-k}.$$

Consequently, $c_k e_k + c_{-k} e_{-k} = 2 \operatorname{Re}(c_k e_k)$ is real. The complex representation organizes a real oscillation into a conjugate pair of frequencies; it does not change the fact that the reconstructed signal is real-valued.

5.3 The best approximation with low frequencies

Keeping finitely many coordinates gives a trigonometric polynomial. We now show that the Fourier coefficients are also the coefficients that minimize its energy error.

Let $\mathcal{E}_N := \operatorname{span}\{e_{-N}, \ldots, e_N\}$ denote the space of trigonometric polynomials $p = \sum_{k=-N}^{N} a_k e_k$. Its angular frequencies have magnitude at most $2\pi N$.

Proposition 5.1. $S_N f$ is the unique minimizer of $\|f - p\|_{L^2}$ over $p \in \mathcal{E}_N$.

Proof. Set $r = f - S_N f$. For every $|\ell| \le N$,

$$\langle r, e_\ell \rangle_{L^2} = c_\ell - \sum_{k=-N}^{N} c_k \langle e_k, e_\ell \rangle_{L^2} = c_\ell - c_\ell = 0.$$

Thus r is orthogonal to every element of $\mathcal{E}_N$. For any $p \in \mathcal{E}_N$, write $f - p = r + (S_N f - p)$ and expand:

$$\begin{aligned} \|f - p\|_{L^2}^2 &= \|r\|_{L^2}^2 + \|S_N f - p\|_{L^2}^2 + 2 \operatorname{Re}\langle r, S_N f - p \rangle_{L^2} \\ &= \|f - S_N f\|_{L^2}^2 + \|S_N f - p\|_{L^2}^2. \end{aligned}$$

The last term is nonnegative and vanishes exactly when $p = S_N f$ in L^2. □

The error decomposition proves more than optimality: the residual is orthogonal to the approximation space. Thus S_N is the *orthogonal projector* onto $\mathcal{E}_N$. In Fourier-series coordinates, this projector is a low-pass filter:

$$S_N e_k = \begin{cases} e_k, & |k| \le N, \\ 0, & |k| > N. \end{cases}$$

It keeps the low frequencies with gain one and removes the rest.

Theorem 5.2 (Fourier series)**.** The family $\{e_k\}_{k\in\mathbb{Z}}$ is a complete orthonormal basis of $L^2[0,1]$. In particular,

$$\boxed{f = \sum_{k\in\mathbb{Z}} \langle f, e_k\rangle_{L^2} e_k \quad \text{in } L^2, \qquad \|f\|_{L^2}^2 = \sum_{k\in\mathbb{Z}} |c_k|^2.} \tag{5.4}$$

The first identity means $\|f - S_N f\|_{L^2} \to 0$; the second is Parseval's identity. Completeness is a standard Fourier-series result and does not come from orthogonality alone.

For a finite partial sum, orthogonality gives

$$\|S_N f\|_{L^2}^2 = \sum_{k=-N}^{N} |c_k|^2, \qquad \|f - S_N f\|_{L^2}^2 = \|f\|_{L^2}^2 - \sum_{k=-N}^{N} |c_k|^2. \tag{5.5}$$

Indeed, expand the first squared norm and cancel the cross terms; then use $f = (f - S_N f) + S_N f$, an orthogonal decomposition. Completeness makes the last error tend to zero and yields the infinite energy identity in equation (5.4).

5.4 Fourier truncation as periodic convolution

The frequency-domain description tells us which coefficients are retained. To understand what the same operation does to the signal itself, insert the coefficient integrals into the finite sum:

$$\begin{aligned} (S_N f)(t) &= \sum_{k=-N}^{N} \left(\int_0^1 f(u) \mathrm{e}^{-\mathrm{j}2\pi k u} \, \mathrm{d}u \right) \mathrm{e}^{\mathrm{j}2\pi k t} \\ &= \int_0^1 f(u) \underbrace{\sum_{k=-N}^{N} \mathrm{e}^{\mathrm{j}2\pi k (t-u)}}_{D_N(t-u)} \, \mathrm{d}u = (f *_{\mathrm{per}} D_N)(t). \end{aligned} \tag{5.6}$$

Definition. $D_N(s) := \sum_{k=-N}^{N} \mathrm{e}^{\mathrm{j}2\pi k s}$ is the *Dirichlet kernel*. The integral in equation (5.6) is *periodic convolution*: arguments outside $[0,1]$ wrap around by periodicity. Unlike convolution on $\mathbb{R}$, we integrate over one period.

Calculate the kernel. For $s \notin \mathbb{Z}$, let $q = \mathrm{e}^{\mathrm{j}2\pi s} \neq 1$. A geometric sum gives

$$D_N(s) = q^{-N} \sum_{r=0}^{2N} q^r = \mathrm{e}^{-\mathrm{j}2\pi N s} \frac{1 - \mathrm{e}^{\mathrm{j}2\pi(2N+1)s}}{1 - \mathrm{e}^{\mathrm{j}2\pi s}}$$
$$= \mathrm{e}^{-\mathrm{j}2\pi N s} \frac{-2\mathrm{j}\mathrm{e}^{\mathrm{j}\pi(2N+1)s} \sin((2N+1)\pi s)}{-2\mathrm{j}\mathrm{e}^{\mathrm{j}\pi s} \sin(\pi s)} = \frac{\sin((2N+1)\pi s)}{\sin(\pi s)}.$$

All phase factors cancel. If $s \in \mathbb{Z}$, use the original sum: $D_N(s) = 2N + 1$.

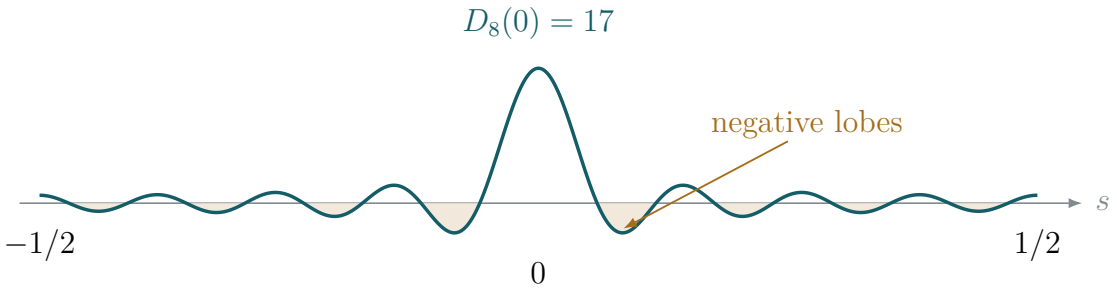


Figure 5.3: The Dirichlet kernel for $N = 8$. Shading marks the negative lobes. Its signed area over one period is one, but the kernel is not nonnegative.

Two checks. Each exponential is 1-periodic, so $D_N(s+1) = D_N(s)$. Also,

$$\int_0^1 D_N(s)\,\mathrm{d}s = \sum_{k=-N}^{N} \int_0^1 \mathrm{e}^{\mathrm{j}2\pi k s}\,\mathrm{d}s = 1.$$

Only $k = 0$ contributes. The kernel is real, since pairing k with $-k$ gives $D_N(s) = 1 + 2\sum_{k=1}^{N} \cos(2\pi k s)$.

The unit integral ensures that constants are reproduced, but it does not make convolution with D_N a positive averaging operation. For $N \geq 1$, the kernel has negative lobes. Their signed contributions can push the approximation outside the range of the original signal, even when $0 \leq f \leq 1$. The step function makes this effect explicit.

5.5 Fourier coefficients of a step

A step lets us calculate the coefficients explicitly and follow the effect of truncation near a discontinuity.

Example. Let $f(t) = \mathbf{1}_{[1/2,1)}(t)$ on $[0,1]$. Then

$$c_0 = \int_{1/2}^{1} 1\,\mathrm{d}t = \frac{1}{2}.$$

For $k \neq 0$,

$$c_k = \int_{1/2}^{1} \mathrm{e}^{-\mathrm{j}2\pi kt}\,\mathrm{d}t = \left[\frac{\mathrm{e}^{-\mathrm{j}2\pi kt}}{-\mathrm{j}2\pi k}\right]_{1/2}^{1}$$
$$= \frac{(-1)^k - 1}{\mathrm{j}2\pi k} = \begin{cases} 0, & k \neq 0 \text{ even}, \\ \dfrac{\mathrm{j}}{\pi k}, & k \text{ odd}. \end{cases}$$

No finite Fourier sum equals this step in L^2. To see this, choose an odd frequency beyond that sum. Its coefficient would be zero by orthogonality, but the coefficient of the step is not zero.

For a positive odd k, combine the positive and negative frequencies:

$$c_k e_k(t) + c_{-k} e_{-k}(t) = \frac{\mathrm{j}}{\pi k}\left(\mathrm{e}^{\mathrm{j}2\pi kt} - \mathrm{e}^{-\mathrm{j}2\pi kt}\right)$$
$$= -\frac{2}{\pi k}\sin(2\pi kt).$$

Set $k = 2r - 1$, for $r = 1, \ldots, M$. Since the even coefficients vanish,

$$\boxed{s_M(t) \coloneqq (S_{2M-1}f)(t) = (S_{2M}f)(t) = \frac{1}{2} - \frac{2}{\pi}\sum_{r=1}^{M} \frac{\sin(2\pi(2r-1)t)}{2r-1}.} \tag{5.7}$$

In this real form, the approximation is specified by $M + 1$ nonzero coefficients: the mean and M sine amplitudes. In the complex Fourier form, each sine uses two conjugate coefficients. The highest angular frequency retained is $2\pi(2M - 1)$.

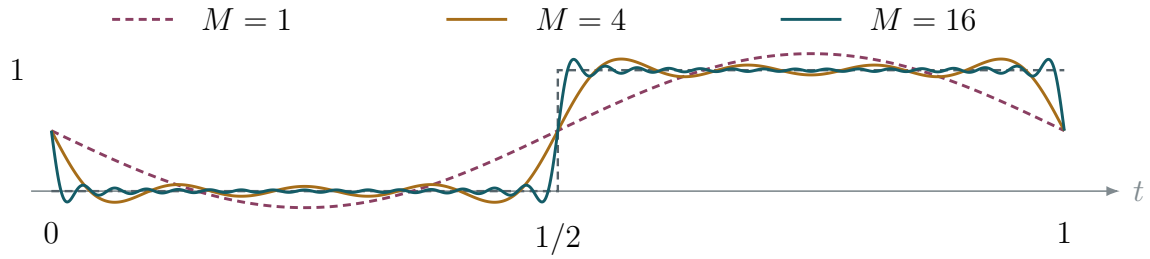


Figure 5.4: The same step with one, four, and sixteen positive odd frequencies. More frequencies sharpen the transition, but ringing remains on both sides of the jump.

5.6 Gibbs overshoot does not go away

The plots suggest that the transition sharpens while the overshoot persists. To distinguish these two effects, we calculate both the location and the height of the first peak. Write $t = 1/2 + x$. Since $2r - 1$ is odd,

$$\sin\big(2\pi(2r-1)(1/2+x)\big) = -\sin\big(2\pi(2r-1)x\big).$$

Thus

$$s_M(1/2+x) = \frac{1}{2} + \frac{2}{\pi}\sum_{r=1}^{M} \frac{\sin(2\pi(2r-1)x)}{2r-1}. \tag{5.8}$$

In particular, $s_M(1/2) = 1/2$ for every M. Differentiate the *finite* sum:

$$\frac{\mathrm{d}}{\mathrm{d}x} s_M(1/2+x) = 4\sum_{r=1}^{M} \cos(2\pi(2r-1)x).$$

To add the cosines, use

$$2\sin\alpha\cos((2r-1)\alpha) = \sin(2r\alpha) - \sin((2r-2)\alpha).$$

The sum telescopes:

$$\begin{aligned} 2\sin\alpha\sum_{r=1}^{M}\cos((2r-1)\alpha) &= [\sin(2\alpha) - \sin 0] + [\sin(4\alpha) - \sin(2\alpha)] \\ &\quad + \cdots + [\sin(2M\alpha) - \sin((2M-2)\alpha)] \\ &= \sin(2M\alpha). \end{aligned}$$

Put $\alpha = 2\pi x$. For $0 < x < 1/2$,

$$\frac{\mathrm{d}}{\mathrm{d}x} s_M(1/2+x) = 2\frac{\sin(4\pi M x)}{\sin(2\pi x)}. \tag{5.9}$$

The denominator is positive. The numerator first changes from positive to negative when $4\pi M x = \pi$. Therefore

$$\boxed{t_M = \frac{1}{2} + \frac{1}{4M}.} \tag{5.10}$$

Calculate the height. Substitute t_M into equation (5.8):

$$s_M(t_M) = \frac{1}{2} + \frac{2}{\pi}\sum_{r=1}^{M} \frac{\sin\big(\pi\frac{2r-1}{2M}\big)}{2r-1}.$$

Set

$$q_r \coloneqq \frac{r-1/2}{M} = \frac{2r-1}{2M}, \qquad 2r-1 = 2Mq_r.$$

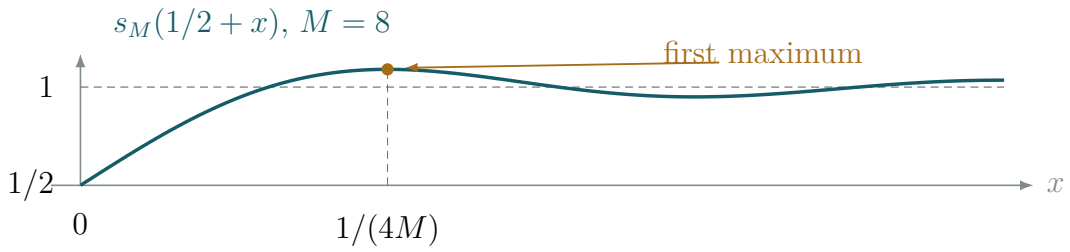


Figure 5.5: Zoom on the first peak: the vertical scale starts at $1/2$. The maximum is exactly $1/(4M)$ to the right of the jump.

These are the midpoints of the M equal subintervals of $[0,1]$. Hence

$$s_M(t_M) = \frac{1}{2} + \frac{1}{\pi} \underbrace{\frac{1}{M}\sum_{r=1}^{M} \frac{\sin(\pi q_r)}{q_r}}_{\text{midpoint Riemann sum on } [0,1]}. \tag{5.11}$$

The function $q \mapsto \sin(\pi q)/q$ extends continuously to $q = 0$ with value π, so the Riemann sum tends to its integral.

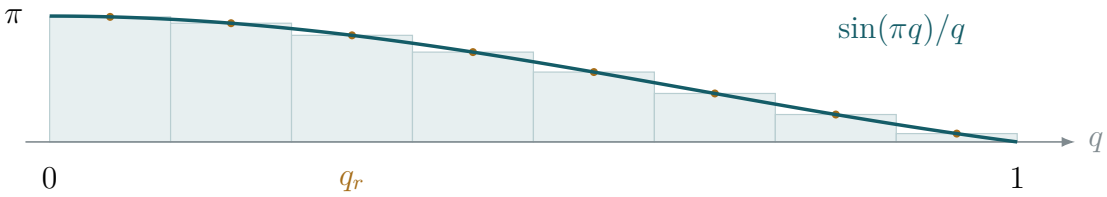


Figure 5.6: The peak height is determined by a midpoint Riemann sum. The apparent singularity at $q = 0$ is removable: the curve starts at height π.

With $v = \pi q$, we have $\mathrm{d}q = \mathrm{d}v/\pi$. Thus

$$\int_0^1 \frac{\sin(\pi q)}{q}\,\mathrm{d}q = \int_0^\pi \frac{\sin v}{v/\pi}\frac{\mathrm{d}v}{\pi} = \int_0^\pi \frac{\sin v}{v}\,\mathrm{d}v.$$

Consequently,

$$\boxed{\lim_{M\to\infty}\big(s_M(t_M) - 1\big) = \frac{1}{\pi}\int_0^\pi \frac{\sin v}{v}\,\mathrm{d}v - \frac{1}{2} \approx 0.089489872.} \tag{5.12}$$

The first peak therefore approaches height $1.08949\ldots$, rather than height 1. Increasing the number of frequencies moves this peak toward the discontinuity, but its excess height approaches about 8.95% of the unit jump. This is the Gibbs phenomenon.

M	distance $t_M - 1/2$	peak $s_M(t_M)$	overshoot
1	1/4	1.136620	13.66%
4	1/16	1.092113	9.21%
16	1/64	1.089653	8.97%

5.7 Small energy is not small error everywhere

The persistent Gibbs overshoot does not contradict convergence in L^2. The L^2 error is an *integral*, not an error bound at every point. Here $\|f\|_{L^2}^2 = 1/2$. The mean contributes $1/4$ of the energy; each pair of odd Fourier coefficients contributes $2/(\pi^2(2r-1)^2)$. By equation (5.5),

$$\|f - s_M\|_{L^2}^2 = \frac{1}{4} - \frac{2}{\pi^2} \sum_{r=1}^{M} \frac{1}{(2r-1)^2}. \tag{5.13}$$

Parseval applied to the full step gives

$$\frac{1}{2} = \frac{1}{4} + \frac{2}{\pi^2} \sum_{r=1}^{\infty} \frac{1}{(2r-1)^2}.$$

Subtracting the retained energy therefore gives

$$\boxed{\|f - s_M\|_{L^2}^2 = \frac{2}{\pi^2} \sum_{r=M+1}^{\infty} \frac{1}{(2r-1)^2} \longrightarrow 0.} \tag{5.14}$$

For example, $(2r-1)^{-2} \le r^{-2}$ and $\sum_{r>M} r^{-2} \le \int_M^\infty x^{-2}\,\mathrm{d}x = 1/M$, so the displayed tail tends to zero. The peak error is evaluated at a point t_M that changes with M and approaches the jump. There is no contradiction: an error can remain appreciable on a shrinking interval while its squared integral tends to zero.

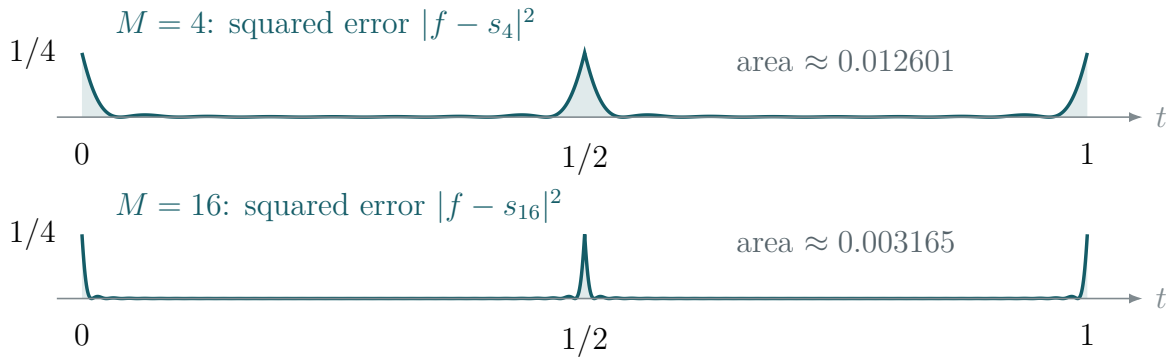


Figure 5.7: The plotted quantity is squared error; its shaded area is the L^2 error squared. The large values occupy narrower regions as M increases. The spikes at the interval endpoints reflect the periodic jump there.

Gibbs phenomenon

The ringing becomes narrower, but not arbitrarily shorter. Its first peak approaches the jump at distance $1/(4M)$, while its excess height tends to $0.08949\ldots$. The total squared error nevertheless tends to zero.

5.8 Fejér averaging

The preceding example identifies a limitation of sharp Fourier truncation, not a prohibition against every Fourier-based approximation. A different choice of weights can suppress the overshoot while keeping the same basis. Fejér averaging provides a particularly transparent example.

Fejér averaging. Average the first $N+1$ partial sums:

$$\begin{aligned} A_N f &:= \frac{1}{N+1} \sum_{m=0}^{N} S_m f \\ &= \frac{1}{N+1} \sum_{m=0}^{N} \sum_{|k|\le m} c_k e_k = \sum_{|k|\le N} \left(1 - \frac{|k|}{N+1}\right) c_k e_k. \end{aligned} \tag{5.15}$$

The coefficient c_k occurs when $m = |k|, \ldots, N$: exactly $N+1-|k|$ times. This explains the triangular weights.

Just as in equation (5.6), $A_N f = f *_{\mathrm{per}} F_N$. The kernel is

$$\begin{aligned} F_N(s) &= \sum_{|k|\le N} \left(1 - \frac{|k|}{N+1}\right) \mathrm{e}^{\mathrm{j}2\pi k s} \\ &= \frac{1}{N+1} \left| \sum_{r=0}^{N} \mathrm{e}^{\mathrm{j}2\pi r s} \right|^2 \ge 0. \end{aligned} \tag{5.16}$$

To verify the second equality, expand the square into $\sum_{r=0}^{N} \sum_{\ell=0}^{N} \mathrm{e}^{\mathrm{j}2\pi(r-\ell)s}$. There are $N+1-|k|$ pairs with $r-\ell = k$. Integrating leaves only the $N+1$ pairs with $r = \ell$, so $\int_0^1 F_N = 1$.

For a real signal with $0 \le f \le 1$, positivity and unit integral imply

$$0 \le (A_N f)(t) = \int_0^1 f(u) F_N(t-u)\,\mathrm{d}u \le \int_0^1 F_N(t-u)\,\mathrm{d}u = 1.$$

Thus Fejér averaging cannot overshoot the range of the step.

This change has a cost. The averaged operator A_N is not the orthogonal projector S_N, since its retained nonzero frequencies have gain strictly below one. It therefore no longer gives the best approximation in the prescribed Fourier subspace.

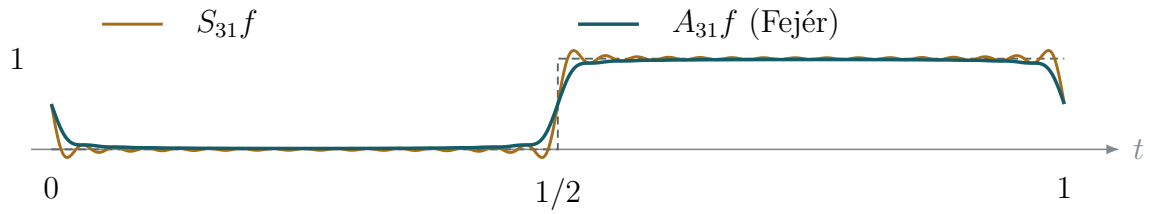


Figure 5.8: Same Fourier frequencies, different weights. Fejér averaging stays within $[0, 1]$, but rounds the transition. The sharp projector is the least-squares approximation; the averaged operator is not.

Moreover, the basis functions are still global: changing a coefficient changes the approximation throughout the interval.

5.9 Localize the representation

To change part of an approximation without changing it everywhere, use building blocks supported on small intervals. In contrast, each Fourier function has $|e_n(t)| = 1$ throughout $[0, 1]$. Changing one Fourier coefficient changes the approximation throughout that interval.

For $i = 0, 1, \dots$ and $n = 0, \dots, 2^i - 1$, define the dyadic cells

$$I_{i,n} := [n2^{-i}, (n+1)2^{-i}), \qquad |I_{i,n}| = 2^{-i}. \tag{5.17}$$

There are 2^i cells. The index i sets their size; n identifies a location. Endpoints can be assigned either value without changing an L^2 function.

Definition. The normalized cell functions are

$$\phi_{i,n}(t) := 2^{i/2}\mathbf{1}_{I_{i,n}}(t) = \begin{cases} 2^{i/2}, & t \in I_{i,n}, \\ 0, & t \notin I_{i,n}. \end{cases} \tag{5.18}$$

The normalization is chosen to preserve energy as the cells become smaller. Indeed, the squared height compensates exactly for the width:

$$\|\phi_{i,n}\|_{L^2}^2 = \int_{I_{i,n}} |2^{i/2}|^2 \, \mathrm{d}t = 2^i |I_{i,n}| = 2^i 2^{-i} = 1.$$

At a *fixed scale* i, two different cells are disjoint, so

$$\langle \phi_{i,n}, \phi_{i,\ell} \rangle_{L^2} = 0 \qquad (n \neq \ell).$$

If g has value b_n on each cell $I_{i,n}$, then

$$g = \sum_{n=0}^{2^i-1} b_n \mathbf{1}_{I_{i,n}} = \sum_{n=0}^{2^i-1} (2^{-i/2} b_n) \phi_{i,n}.$$

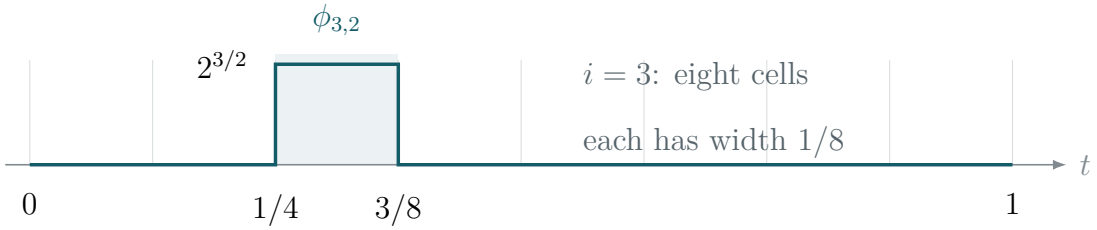


Figure 5.9: At a fixed scale, the normalized indicators have disjoint nonzero regions. Changing the coefficient of $\phi_{3,2}$ changes only the highlighted cell.

This representation proves completeness within the space of signals that are constant on these cells. Together with the orthonormality calculation, it shows that the normalized cell functions form an orthonormal basis for that space. This statement concerns one fixed scale i; functions from different scales need not be orthogonal.

Calculate the local projection. As in Section 5.3, project onto the span of the orthonormal family:

$$P_i f = \sum_{n=0}^{2^i-1} \langle f, \phi_{i,n} \rangle_{L^2} \phi_{i,n}. \tag{5.19}$$

Since the cell functions are real,

$$\langle f, \phi_{i,n} \rangle_{L^2} = \int_0^1 f(u)\phi_{i,n}(u)\,\mathrm{d}u = 2^{i/2} \int_{I_{i,n}} f(u)\,\mathrm{d}u.$$

For $t \in I_{i,n}$, all terms except the nth vanish. Therefore

$$\begin{aligned}(P_i f)(t) &= \left(2^{i/2} \int_{I_{i,n}} f(u)\,\mathrm{d}u\right) 2^{i/2} \\ &= 2^i \int_{I_{i,n}} f(u)\,\mathrm{d}u = \boxed{\frac{1}{|I_{i,n}|} \int_{I_{i,n}} f(u)\,\mathrm{d}u.}\end{aligned} \tag{5.20}$$

Thus the orthogonal projection replaces the signal on each cell by its *average on that cell*. The construction is the same Hilbert-space projection used for Fourier truncation, but the basis functions now have local support. Consequently, the value of the approximation on a cell depends only on the signal on that cell.

Example. Move the jump to 1/3: let $f_{1/3} = \mathbf{1}_{[1/3,1)}$. At $i = 3$, the jump lies in $I_{3,2} = [1/4, 3/8)$. The value in this cell is

$$8 \int_{1/4}^{3/8} f_{1/3}(u)\,\mathrm{d}u = 8\left(\frac{3}{8} - \frac{1}{3}\right) = \frac{1}{3}.$$

All cells to its left have average zero; all cells to its right have average one.

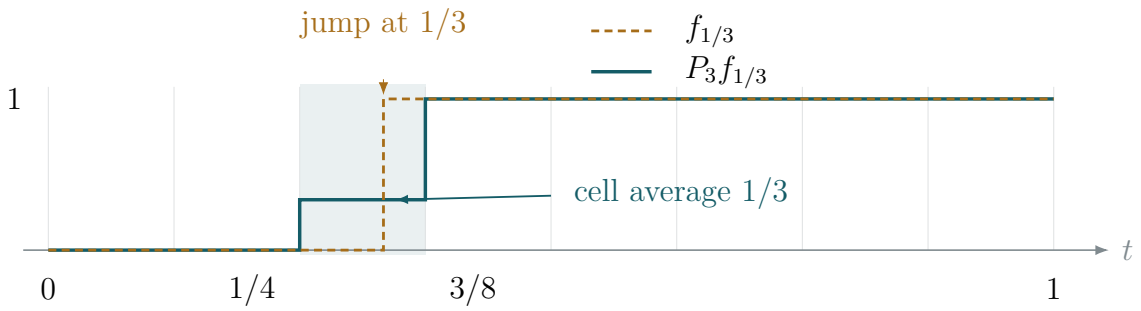


Figure 5.10: Local averaging replaces the portion of the step inside $[1/4, 3/8)$ by its mean. All the approximation error is confined to that one cell.

Only the cell containing the jump has an approximation error. No oscillatory correction is spread across the interval. If $0 \le f \le 1$, every cell average also lies between zero and one.

The midpoint step from Section 5.5 is even simpler: it is already constant on the two cells at $i = 1$, so $P_1 f = f$. This suggests replacing global sinusoids by local averages and local differences.

5.10 From local averages to wavelets

Definition. The Haar *scaling function* and *wavelet* are

$$\phi(t) := \mathbf{1}_{[0,1)}(t), \qquad \psi(t) := \mathbf{1}_{[0,1/2)}(t) - \mathbf{1}_{[1/2,1)}(t).$$

The box ϕ is also the zeroth-order B-spline. It represents the common level on an interval. The wavelet is positive on its left half and negative on its right half, and therefore measures a contrast between the two halves. These are the continuous-domain counterparts of an average and a difference. Together they separate the two half-cell values into a common level and the change about that level.

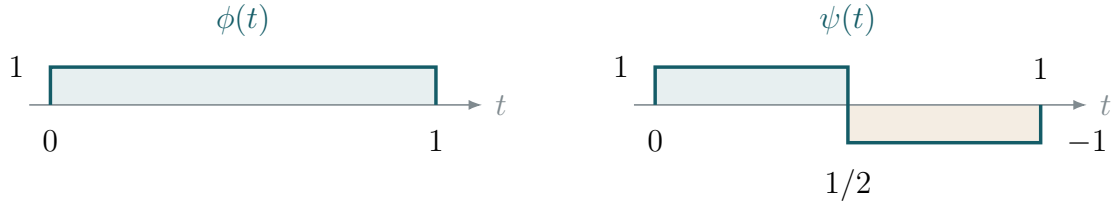


Figure 5.11: An average and a contrast. The wavelet has equal positive and negative areas. Vertical segments mark jumps, not extra values of the function.

Since $\int_0^1 \psi(t)\,\mathrm{d}t = 1/2 - 1/2 = 0$, the positive and negative areas cancel. Both functions have energy one:

$$\|\phi\|_{L^2}^2 = 1, \qquad \|\psi\|_{L^2}^2 = \frac{1}{2} + \frac{1}{2} = 1, \qquad \langle \phi, \psi \rangle_{L^2} = \int_0^1 \psi(t)\,\mathrm{d}t = 0.$$

Thus ϕ, ψ are an orthonormal pair.

Example. Return to $f = \mathbf{1}_{[1/2,1)}$. Its two coefficients are

$$\langle f, \phi \rangle_{L^2} = \int_{1/2}^1 1\,\mathrm{d}t = \frac{1}{2}, \qquad \langle f, \psi \rangle_{L^2} = \int_{1/2}^1 (-1)\,\mathrm{d}t = -\frac{1}{2}.$$

$$\boxed{f = \frac{1}{2}\phi - \frac{1}{2}\psi.} \tag{5.21}$$

On the left half this is $1/2 - 1/2 = 0$; on the right it is $1/2 + 1/2 = 1$. Two real coordinates reconstruct the step exactly, without overshoot.

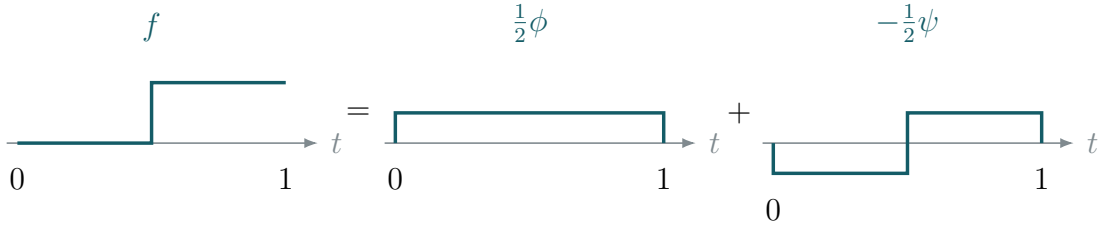


Figure 5.12: The step is a constant background plus a signed local contrast. Unlike a finite trigonometric sum, these two functions reproduce the jump exactly.

The two-term representation uses the fact that the jump aligns with the midpoint. A jump at another location generally requires additional wavelets at finer scales. The useful principle is not that every step has two coefficients, but that a localized change can be represented using localized functions near that change.

Examine smaller regions. For $i \geq 0$ and $0 \leq n < 2^i$, define

$$\boxed{\psi_{i,n}(t) := 2^{i/2}\psi(2^i t - n).} \tag{5.22}$$

Since $0 \leq 2^i t - n < 1$ exactly when $t \in I_{i,n}$, each wavelet occupies one cell. Increasing i makes it narrower; changing n moves it. Its two halves have opposite signs.

Check the normalization. Put $u = 2^i t - n$, so $\mathrm{d}t = 2^{-i}\mathrm{d}u$. Since ψ vanishes outside $[0, 1)$,

$$\|\psi_{i,n}\|_{L^2}^2 = 2^i 2^{-i} \int_0^1 |\psi(u)|^2\,\mathrm{d}u = 1,$$
$$\int_0^1 \psi_{i,n}(t)\,\mathrm{d}t = 2^{i/2}2^{-i} \int_0^1 \psi(u)\,\mathrm{d}u = 0.$$

Narrower wavelets are taller, but every wavelet has the same energy and zero mean.

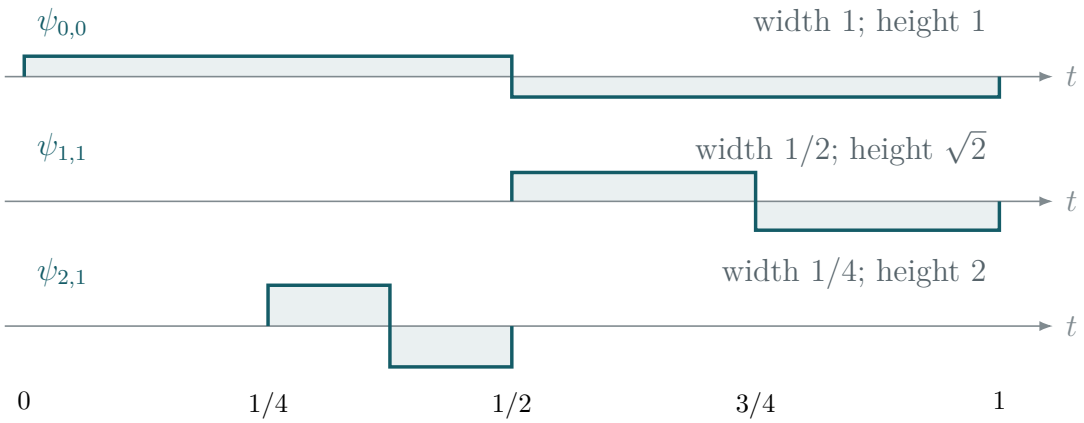


Figure 5.13: The same Haar pattern at different positions and scales, with a common horizontal axis. The amplitude factor keeps the energy equal to one.

Cancellation on constant regions. If $f(t) = b$ on $I_{i,n}$, then

$$\langle f, \psi_{i,n} \rangle_{L^2} = b \int_{I_{i,n}} \psi_{i,n}(t)\, \mathrm{d}t = 0.$$

A constant background cancels; a local contrast need not.

The need for localization

Fourier functions describe global frequency content. Haar wavelets describe local changes at different scales. Their coefficients let us place representational detail where the signal changes, rather than alter the entire interval with each coordinate. This is the starting point for multiresolution signal analysis.